
Hedge Fund Trading and Sovereign Bond Yield Sensitivity

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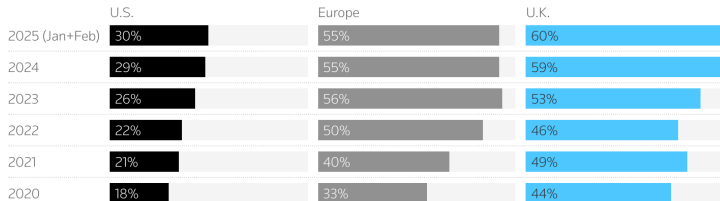
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Hedge funds hold a growing footprint in government bond markets

Hedge funds in govies

Notional percentage volumes in government bonds tracked by Tradeweb

● U.S. ● Europe ● U.K.



Source: Tradeweb | Chart by Nell Mackenzie

- Their share of trading volume keeps rising since 2020
- In Europe it went from a third to more than half of volumes
- The rise is broad, across the US, Europe, and the UK

► Do hedge funds change how prices incorporate new information?

Two views on the impact of leveraged investors on price

Decrease sensitivity

Their arbitrage activity corrects mispricings
Prices are pushed *toward* fundamentals

Gromb and Vayanos (2002) and Cao et al. (2018)

Increase sensitivity

Leverage and funding fragility force trades
Prices are pushed *away* from fundamentals

Shleifer and Vishny (1997), Brunnermeier and Pedersen (2009), and
Geanakoplos (2010)

Research question

- Does the **leveraged positioning** of hedge funds amplify sovereign bond yield sensitivity to **shocks**, and under what conditions?

Hedge fund repo positions ECB SFTDS

- Regulatory transaction data covering euro area repo activity at the transaction level
- German and Italian sovereign bonds, Jan 2021 to Nov 2025 (about 70% of euro area hedge fund sovereign activity)
- 152 identified hedge funds

Bond characteristics Bloomberg

- Daily yields, duration, bid ask spread, amount issued, cheapest to deliver status

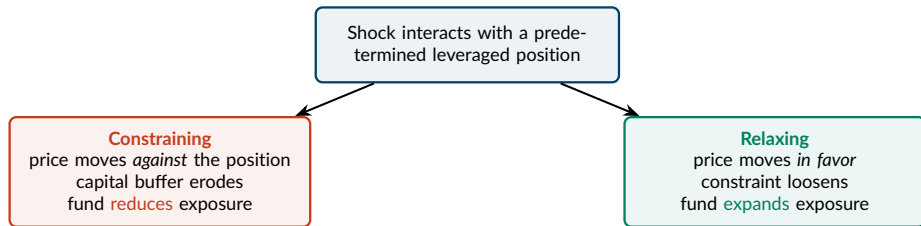
Exogenous shocks EA-MPD (Altavilla et al. [2019](#))

- Intraday change in the 2 year OIS rate during the ECB monetary event window, the same shock hits every bond, so identification comes from cross sectional differences in positioning

How leveraged positioning amplifies yields

Starting point Adrian and Shin (2014)

- Leveraged intermediaries actively manage their balance sheet against the risk environment
- The same rule produces **opposite** position adjustments depending on the sign of the shock



► In both states the resulting trading pushes yields further in the direction of the initial shock

Baseline panel regression

$$\Delta y_{i,t} = \beta_1 (\text{HF}_{i,t-1} \times \text{Shock}_t) + \beta_2 \text{HF}_{i,t-1} + \beta_3 \text{Shock}_t + \gamma X_{i,t-1} + \theta_k + \varepsilon_{i,t}$$

Positioning and β_1

- **Involved**, a binary indicator of an active position, and **intensity**, the five day average net position scaled by amount outstanding
- β_1 is the additional yield sensitivity to the shock when hedge funds are active

Two challenges to a causal reading

- Hedge funds **select** liquid, large, longer duration, and cheapest to deliver bonds
- Duration $\times$ country $\times$ date fixed effects compare similar bonds from the same issuer on the same day, and a matched design pairs each bond with its closest twin
- A dealer level panel shows the trigger is **not** dealer behavior

Finding 1: Hedge funds increase the volatility of bond yields on shock days

$ \Delta \text{ yield} $ (bps)	(1) Raw comparison	(2) Comparable bonds
HF involved $\times$ Shock	0.240***	0.246***
HF involved (level)	0.459***	−0.098
Shock	0.545***	
Bond controls	No	Yes
Bond FE	No	Yes
Duration $\times$ Date $\times$ Country FE	No	Yes

Dependent variable is the absolute daily yield change. Column 2 identifies the effect within cells of bonds with similar duration from the same issuer on the same day. Intermediate specifications in the annex. Standard errors clustered by date and bond. *** $p < 0.01$.

In the raw comparison hedge fund bonds are more volatile on **every** day, hedge funds select volatile bonds.

Among comparable bonds the **level** differential disappears while the **interaction** with the shock survives.

► The excess volatility is shock conditional, so what changes is how yields react to shocks

Finding 2: Hedge fund trading raises the sensitivity of yields to shocks

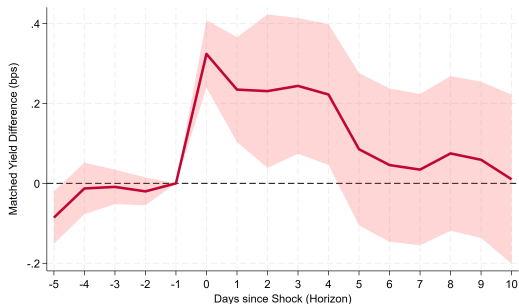
	Presence		Intensity
Δ yield (bps)	(1)	(2) Matched	(3)
HF involved $\times$ Shock	0.291***	0.304***	
HF intensity $\times$ Shock			0.164***
HF involved (level)	-0.022	-0.002	
HF intensity (level)			-0.011
Bond controls	Yes	Yes	Yes
Bond FE	Yes	No	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	No	Yes
Matched pair FE	No	Yes	No

Only the two tightest designs and the continuous dose response, all weaker specifications in the annex. HF intensity is the log of one plus the five day net position scaled by amount outstanding. Column 2 pairs each hedge fund bond with the twin closest in liquidity within its cell. Standard errors clustered by date and bond, and by cell and twin in column 2.

*** $p < 0.01$.

Presence and intensity point the same way. Within the **same bond** over time, a larger position produces a larger reaction, which pure selection cannot explain.

Yields diverge after a shock

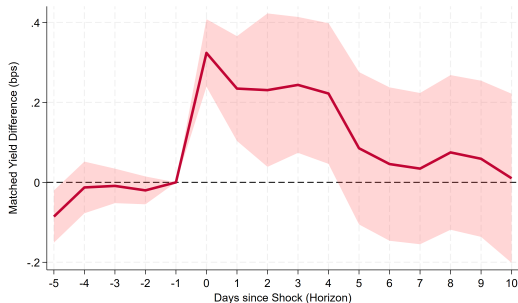


(a) Matched differential

Local projections on the matched sample (Jordà 2005). Shaded areas are 90% confidence intervals clustered by date and bond.

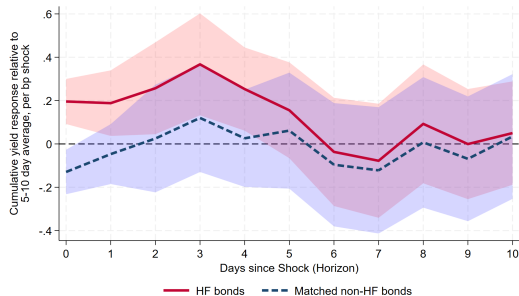
Flat before the shock, immediate divergence on impact,
significant for about **four days**, then reversion.

Yields diverge after a shock, reflecting temporary overshooting



(a) Matched differential

Local projections on the matched sample (Jordà 2005). Shaded areas are 90% confidence intervals clustered by date and bond.



(b) Level decomposition

Flat before the shock, immediate divergence on impact, significant for about **four days**, then reversion.

► **A temporary dislocation that other capital gradually absorbs, consistent with slow moving capital**

Both legs converge to a common level. Hedge fund bonds **overshoot** first.

Not price discovery

Finding 3: The amplification holds whether shocks help or hurt the position

The same **intensity** specification as before, now estimated separately by regime.

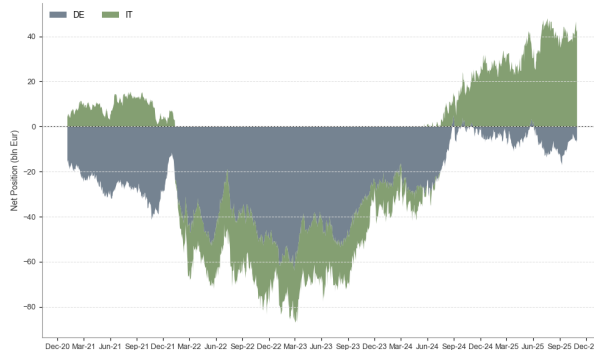
Δ yield (bps)	Constraining	Relaxing
HF int. $\times$ Shock	0.140***	0.224***
HF int. (level)	-0.016	-0.009
Bond controls	Yes	Yes
Bond FE	Yes	Yes
Dur. $\times$ Date $\times$ Ctry FE	Yes	Yes

Each regime also includes all non shock days and zero intensity days as the within bond baseline. Standard errors clustered by date and bond.

- **Constraining** when the shock moves against the position
- **Relaxing** when it moves in favor
- A Wald test **cannot distinguish** the two ($p = 0.16$)
- Positions indeed **contract and expand** after the shock, as the mechanism predicts

► **Amplification is not only a crisis phenomenon, it operates whenever shocks meet positions, consistent with rebalancing toward a target**

Finding 4: The directionality of the book governs the strength



Aggregate net repo positions of offshore hedge funds in German and Italian sovereign bonds, EUR outstanding. Positive values are net cash borrowing (a proxy for long), negative values net cash lending (a proxy for short). Sample period 2021 to 2025.

► So the directionality of the book matters, beyond how much hedge funds hold

Bond level holder directionality

$$\text{Dir}_{f,t} = \frac{|\sum_i D_i q_{f,i,t}|}{\sum_i D_i |q_{f,i,t}|}$$

$$\text{HolderDir}_{i,t} = \frac{\sum_f |q_{f,i,t}| \text{Dir}_{f,t}}{\sum_f |q_{f,i,t}|}$$

Zero when long and short legs offset in duration, close to one

when the book is one sided. Each bond inherits the position weighted directionality of the funds holding it.

Δ yield (bps), relative to no HF	Coefficient
Hedged book $\times$ Shock	0.098***
Directional book $\times$ Shock	0.363***
<i>By directionality quartile</i>	
Q1 most hedged $\times$ Shock	0.057
Q2 $\times$ Shock	0.137***
Q3 $\times$ Shock	0.361***
Q4 most directional $\times$ Shock	0.367***

Bond and duration $\times$ country $\times$ date fixed effects with controls. Wald tests reject equality of the hedged and directional groups ($F = 24.8$) and of the extreme quartiles ($F = 28.3$). Standard errors clustered by date and bond. *** $p < 0.01$.

- The amplification rises monotonically with directionality and is not distinguishable from zero in the most hedged quartile

What the paper shows

1. Hedge funds select more volatile bonds, but the excess volatility is limited to shock days
2. Their trading raises the sensitivity of yields to shocks, more than a quarter of the shock day move, and the overshoot reverts within days
3. The amplification is symmetric across the constraining and relaxing states, consistent with constant leverage targeting
4. It is concentrated in **directional** books and close to zero in **hedged** books

For risk monitoring

- Size and presence alone are not sufficient, and amplification is not confined to stress episodes
- The **directionality** of the book is a key risk metric beyond position size

Thank you

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Annex

- **Determinants of sovereign yields and non fundamental pricing**

I document a specific source of temporary deviations from fundamentals

De Grauwe and Ji ([2013](#)); Greenwood and Vayanos ([2014](#)); Kojien et al. ([2021](#)); Duffie ([2010](#))

- **Nonbanks and hedge funds in financial markets**

I show an amplification channel that is not confined to stress episodes

Barth and Kahn ([2025](#)); Kruttli et al. ([2025](#)); Anaya et al. ([2025](#)); Pelizzon et al. ([2025](#))

- **How leveraged investors transmit shocks**

I show comparable amplification in the constraining and the relaxing state

Adrian and Shin ([2014](#)); Brunnermeier and Pedersen ([2009](#)); Avalos and Sushko ([2023](#)); De Long et al. ([1990](#)); Moskowitz et al. ([2012](#))

Summary statistics

	Mean	SD	P25	Median	P75	N
<i>Bond market</i>						
Daily yield change (bps)	0.15	5.79	-2.25	0.00	2.75	477,001
Duration (years)	7.13	7.00	1.77	5.14	10.47	477,001
Bid ask spread (€)	0.31	0.42	0.03	0.11	0.44	477,001
<i>Hedge fund positioning</i>						
HF present (dummy)	0.53	0.50	0.00	1.00	1.00	477,001
Net position, active (% issued)	3.01	3.29	0.64	1.80	4.26	253,111
<i>Repo terms, active</i>						
Haircut, long (%)	0.24	0.54	0.00	0.12	0.30	109,068
Maturity, long (days)	16.49	16.42	7.50	11.00	18.65	101,535
<i>Shock days</i>						
Tightening surprise (bps)	5.87	4.10	3.30	4.95	7.60	5,736
Easing surprise (bps)	-4.17	4.48	-6.72	-3.13	-0.56	8,803

Daily panel of German and Italian sovereign bonds. Positioning variables are five day moving averages scaled by amount outstanding. Sample period 2021 to 2025.

A numerical example

Setup representative sample values

- A bond of duration about 7 years financed via repo, effective margin $m_i = 2\%$ (raw haircuts 0.24%)
- Hedge funds hold $Q_{HF} / Q_{Total} = 3\%$ of the bond
- Tightening surprise of 6 bps, market elasticity $\lambda = 5$ bps per pp traded (D'Amico and King 2013; Eser and Schwaab 2016; De Santis and Holm-Hadulla 2020)

Step 1 the shock erodes the capital buffer

$$\frac{\Delta E}{E} \approx \frac{1}{m_i} \times \frac{\Delta P}{P} = \frac{-0.42\%}{2\%} = -21\%$$

Step 2 restoring leverage generates trading pressure

$$21\% \times 3\% = 0.63\% \text{ of the amount outstanding}$$

Step 3 trading pressure moves the yield

$$\Delta y^{\text{Amp}} = \lambda \times 0.63 = 5 \times 0.63 = 3.15 \text{ bps}$$

Hedge funds do not pick bonds at random

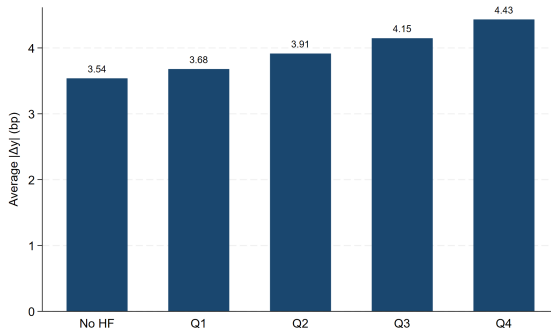
HF involvement	(1)	(2)	(3)
Duration	0.012***	0.012***	0.013***
Bid ask spread	-0.534***	-0.524***	-0.549***
Amount issued	0.030***	0.032***	0.032***
CTD flag	0.070**	0.070**	0.070**
Country FE	No	Yes	No
Country $\times$ Date FE	No	No	Yes
R^2	0.518	0.533	0.543

Linear probability model, dependent variable is a binary indicator for offshore hedge fund repo activity. $N = 477,001$. Standard errors clustered by date and bond. *** $p < 0.01$, ** $p < 0.05$.

Hedge funds tilt toward **liquid**, **larger**, longer duration, and cheapest to deliver bonds, which could move shock sensitivity on their own.

► A simple correlation would confound leveraged capital with the bonds hedge funds prefer

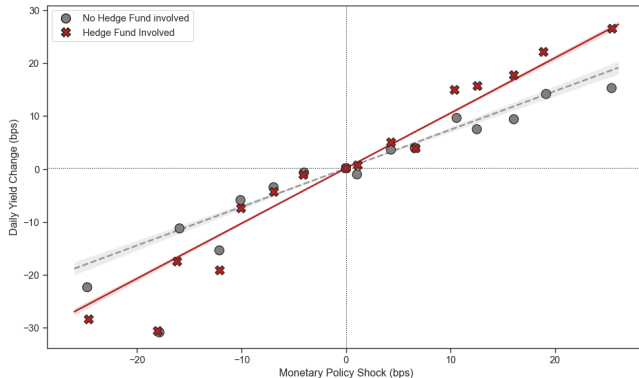
Raw volatility by hedge fund intensity



Average absolute daily yield change in basis points on days without a monetary policy shock. Buckets are quartiles of hedge fund intensity, the five day net position scaled by amount outstanding, among bonds with active positions, next to a no hedge fund reference. Raw group means without controls. Sample period 2021 to 2025.

► Unconditionally, hedge funds sit in more volatile bonds

The reaction to the same surprise, with and without hedge funds



Binned scatterplot of daily yield changes against monetary policy shocks, separately for bonds with and without active hedge fund positions. Yield changes are residualized for bond fixed effects. Lines are linear fits on the bond level data. Sample period 2021 to 2025.

► The same surprise moves yields visibly more where hedge funds are positioned

Back

The trigger is hedge fund positioning, not dealer behavior

Δ yield (bps)	(1)	(2)	(3)	(4)
HF involved $\times$ Shock	0.271***	0.294***	0.248***	0.250***
HF involved (level)	0.006	0.007*	0.001	0.001
HF $\times$ Shock $\times$ Δ CDS				0.000
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dealer $\times$ Date FE	Yes	No	No	No
Dealer $\times$ Date $\times$ Country FE	No	Yes	No	No
Dealer $\times$ Date $\times$ Dur. FE	No	No	Yes	Yes

Unit of observation is dealer, bond, day, balanced with zeros. Δ CDS is the dealer's own five day CDS change before the shock. Standard errors clustered by dealer, date, and bond.

Comparing bonds traded by the **same dealer on the same day**, the hedge fund bonds still react more. The effect does **not** scale with the dealer's own credit stress.

► **Dealers intermediate the flow, but the trigger is the leveraged positioning of the funds**

The amplification is not price discovery

A price discovery story predicts **stronger** amplification when the surprise carries new fundamental information.

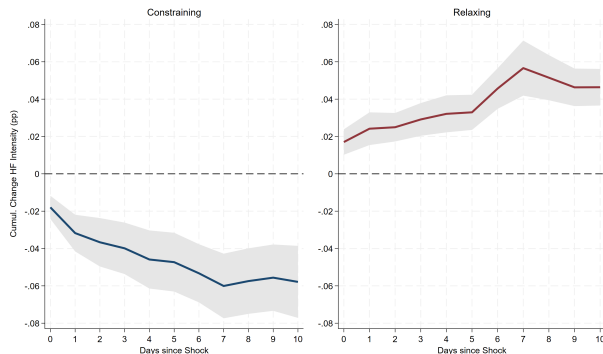
Δ yield (bps)	(1) Pure monetary	(2) Information
HF involved $\times$ Shock	0.377***	0.113***
HF involved (level)	-0.019	-0.026
Bond controls	Yes	Yes
Bond FE	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes

Both columns include bond and duration $\times$ country $\times$ date fixed effects and controls. A Wald test rejects equality ($p = 0.065$). Standard errors clustered by date and bond.

The amplification is concentrated in the **pure monetary** component, the part that carries **no** new fundamental information. This is the **opposite** of the price discovery prediction.

► **Decomposing the surprise points the same way as the overshooting** (Jarociński and Karadi 2020)

Positions contract and expand as the mechanism predicts



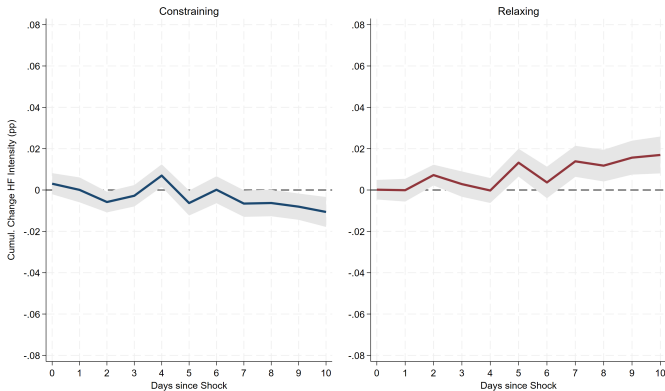
Cumulative change in hedge fund intensity from $t - 1$ to $t + h$ after a shock, by regime, local projections on overnight repo (Jordà 2005). Bond and duration $\times$ country $\times$ date fixed effects. Shaded areas are 90% confidence intervals. Sample period 2021 to 2025.

► Consistent with rebalancing toward a target

Term repo

Back

Term repo shows the same pattern with a lag



Cumulative change in hedge fund intensity by regime, term repo only, local projections (Jordà 2005). Same contraction and expansion as overnight, delayed by about five days in line with the maturity profile. Sample period 2021 to 2025.

External validity, expanded sovereign sample

Δ yield (bps)	(1) Presence	(2) Intensity	(3) Constraining	(4) Relaxing
HF involved $\times$ Shock	0.192***			
Log HF intensity $\times$ Shock		0.122***	0.107***	0.171***
HF involved (level)	-0.046			
Log HF intensity (level)		-0.015	-0.019	-0.012
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes	Yes	Yes

Adds French and Spanish sovereigns to the German and Italian baseline. Bond and duration $\times$ country $\times$ date fixed effects with controls. Standard errors clustered by date and bond. *** $p < 0.01$.

Germany, Italy, France, and Spain. All coefficients stay positive and significant, somewhat smaller, consistent with hedge funds being less active in French and Spanish bonds.

Amplification under a sovereign credit shock

Δ yield (bps)	(1)	(2)	(3) Constraining	(4) Relaxing
HF involved $\times$ Shock	0.225***			
Log HF intensity $\times$ Shock		0.107***	0.129***	0.117***
HF involved (level)	−0.030			
Log HF intensity (level)		−0.021	−0.022	−0.020
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes	Yes	Yes

Standard errors clustered by date and bond. *** $p < 0.01$.

The shock is the residual of the daily change in the five year US dollar Italian sovereign CDS spread, restricted to its 2% tails, away from ECB events. The two regimes are close in magnitude.

Volatility standardized yield changes

Standardized Δ yield (σ units)	(1)	(2)	(3) Constraining	(4) Relaxing
HF involved $\times$ Shock	0.043***			
Log HF intensity $\times$ Shock		0.020***	0.016***	0.031***
HF involved (level)	−0.007			
Log HF intensity (level)		−0.004	−0.005	−0.004
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes	Yes	Yes

Standard errors clustered by date and bond. *** $p < 0.01$.

Each yield change is divided by the bond's own conditional volatility from a GARCH(1,1) fit on non shock days. With average $\hat{\sigma}$ near 6 bps these map back to roughly 0.26 and 0.12 bps per bp, close to the baseline.

Placebo with shifted shock dates

Δ yield (bps)	(1)	(2)	(3) Constraining	(4) Relaxing
HF involved $\times$ Placebo shock	−0.030			
Log HF intensity $\times$ Placebo shock		−0.018	−0.014	−0.028
Placebo shock	0.007	0.004	0.002	−0.003
HF involved (level)	−0.031			
Log HF intensity (level)		−0.018	−0.017	−0.020
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes	Yes	Yes

Monetary shock dates shifted by 15 trading days, which breaks the timing. Standard errors clustered by date and bond.

Shifting the shock dates removes the coincidence with announcements. The amplification disappears entirely, which ties it to the actual shock.